Offline — Training happens once

Kaggle Hotel Reservations CSV
This is the downloaded dataset used to train the machine-learning models. It contains information about hotel bookings, such as guest details, room type, lead time, booking status, and room price.

Data Cleaning & Feature Preparation
The data is checked for missing, incorrect, or unnecessary values. Categorical values are converted into a format that the machine-learning models can understand.

Train & Evaluate Two Models
Two models are trained and tested. The classification model predicts whether a booking will be cancelled, while the regression model predicts the average room price.

Saved Model Files
The trained preprocessing pipeline and the two best models are saved as `.pkl` files. These files are later used by the live application, so the models do not need to be trained again.

Runtime — Live Application

User
The user opens the dashboard and enters information about a hotel reservation.

React Dashboard deployed on Vercel
This is the frontend that the user sees in the browser. It collects the booking details, sends them to the backend, and displays the prediction results.

FastAPI `/predict`
This is the backend prediction endpoint. It receives the booking values from the React dashboard, loads the saved preprocessing pipeline and models, and performs the predictions.

Inference Result
This is the final output returned by the backend. It contains the predicted cancellation probability and the predicted room price.

Connection between training and runtime

The saved model files are created during offline training and loaded by FastAPI when the application starts. When the user submits booking details, React sends an HTTPS request to FastAPI, which runs inference and sends the results back to the dashboard. No model training happens while the user is using the application.